

Hints, Placebos and Shortcuts: Controlled Tests of Structural Primers for LLM Graph Reasoning

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Abstract

Prepending a *primer* of factual sentences about a graph’s structure (node degree, clustering coefficient, random-walk return probabilities or component count) to its text encoding is a natural way to assist a language model on graph-reasoning questions. An apparent gain is uninformative if the answer can be read from the primer without consulting the graph, and confounded if the primer merely lengthens the prompt. We evaluate primers against four controls jointly: a graph-blind shortcut solver refit at the evaluated graph size, a length-matched content-free filler, a minimal-length content control, and a degree-preserving rewiring that varies triangle count at fixed prompt length. Across four Qwen3 arms (1.7B and 4B, with and without thinking mode) on 40-node Erdős–Rényi graphs, primers that state the answer help a model at floor and harm a model at ceiling (cross-fitted interaction with baseline accuracy -11.0 points, 95% CI $[-17.1, -5.7]$), whereas primers that do not state it show no such relation. The pattern replicates on eleven held-out arms (0.6–14B, two families) but is not graded at intermediate baselines. The only effect of a primer that does not state the answer, `clustering on node_degree`, is small relative to no primer ($+1.9$ points) and confined to intermediate densities; two candidate mechanisms for it are not supported.

1 Introduction

Fatemi et al. (2023) show that a large language model’s accuracy on graph-reasoning questions varies substantially with the choice of text encoding when the graph and question are held fixed. We ask a complementary question about the same benchmark: does prepending a short *primer* of derived, node-level facts to the graph encoding improve the model’s answers?

Four confounds complicate the interpretation of such an effect. First, a primer may help because

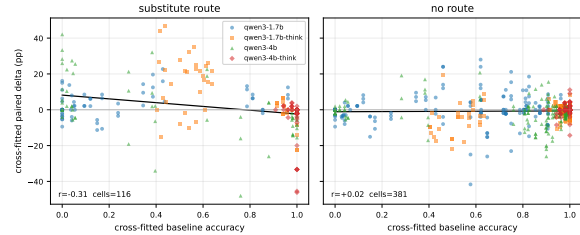


Figure 1: Cross-fitted paired effect against cross-fitted baseline accuracy (Section 5.4) for the 116 cells in which a primer offers a substitute route to the answer (left) and the 381 that do not (right). Each cell appears once per fold direction. Because baseline and effect are estimated from disjoint halves of the paired graphs, shared sampling noise cannot produce the slope; the bounded accuracy scale can, which Section 5.4 examines separately.

the model reads the answer from it rather than reasoning over the graph (answer leakage). Second, a primer changes prompt length, and context length and position alone affect retrieval and reasoning accuracy (Liu et al., 2024; Kamradt, 2023; Levy et al., 2024). Third, answer extraction can err in ways that correlate with response style, which appears as a spurious condition effect (Section 3.4). Fourth, a model near ceiling on a task leaves no room for any primer effect to appear.

We address three research questions. **RQ1:** when a primer improves accuracy, is the improvement attributable to the primer stating the answer? **RQ2:** does added text affect accuracy independently of its content? **RQ3:** does any primer that does not state the answer improve accuracy beyond a length-matched control, and through what mechanism? Our contributions are (i) a battery of controls applied jointly: a graph-blind shortcut solver refit at the evaluated graph size (Section 3.2), a length-matched content-free filler, a minimal-length content control (`components`), and a degree-preserving rewiring that changes triangle count at fixed prompt length (Section 3.3);

(ii) an empirical map of primer effects on 40-node Erdős–Rényi graphs across four model arms and six tasks; (iii) evidence that answer-stating primers act as a substitute route whose value depends on the model’s own baseline accuracy, supported by a cross-fitted interaction test and replicated on held-out models and under node renaming; and (iv) a characterisation of the single effect of a primer that does not state the answer, together with two mechanism tests that it fails.

2 Related Work

We adopt the GraphQA task definitions of [Fatemi et al. \(2023\)](#) and manipulate graph size and edge density directly, so that difficulty is an experimental variable rather than an incidental one. [Wang et al. \(2023\)](#) report that LLMs are brittle to spurious correlations in graph problems and propose *Build-a-Graph* and *Algorithmic* prompting, the prompt-level interventions closest to a primer; [Perozzi et al. \(2024\)](#) instead learn a continuous graph encoding. [Guo et al. \(2023\)](#) find that models struggle to exploit graph description languages directly, and [Jin et al. \(2024\)](#) survey the wider literature. [Dziri et al. \(2023\)](#) argue that transformers solve compositional tasks by linearised subgraph matching rather than systematic reasoning, which raises the same question as a primer’s benefit: does it reflect reasoning or a shorter pattern to match? [Geirhos et al. \(2020\)](#) frame this as *shortcut learning*; our shortcut solver operationalises that concern for the primer channel. Irrelevant context degrades reasoning ([Shi et al., 2023](#)), as does input length well below the context limit ([Levy et al., 2024](#)); our *filler* condition measures this cost directly. The *rwse* condition adapts the random-walk structural encoding of [Dwivedi et al. \(2022\)](#), rendered as sentences, and the rewiring follows the double-edge-swap configuration model ([Fosdick et al., 2018](#)). For statistics we follow [Dror et al. \(2018\)](#) on paired testing, test differences between groups directly rather than contrasting their significance ([Gelman and Stern, 2006](#)), and borrow cross-fitting from [Chernozhukov et al. \(2018\)](#). We distinguish Qwen3’s native thinking mode, toggled through the chat template, from chain-of-thought *prompting* ([Wei et al., 2022](#)); this paper tests only the former.

3 Method

3.1 Primers

A primer is a preamble of factual sentences about a graph’s structure, prepended to the standard text encoding and the task question (a complete example appears in Appendix A.1). A single renderer produces all seven conditions, so conditions share sentence framing and number formatting and differ only in the facts stated. The exception is **filler**, which states no fact and therefore uses its own frame (“Node N is simply present within the graph G .”); it is a length control, not a content control. **none** is the graph encoding alone. **components** gives one sentence stating the number of connected components. **degree** states each node’s degree, **clustering** each node’s clustering coefficient, and **rwse** each node’s random-walk return probabilities after two and three steps ([Dwivedi et al., 2022](#)). **all** combines degree, clustering and *rwse*. Because the model reads tokens rather than characters, we measure length with the Qwen3 tokenizer: mean primer length at $n=40$ is 8 tokens (*components*), 372 (*degree*), 470 (*filler*), 510 (*clustering*), 1,070 (*rwse*) and 1,612 (*all*), against a mean *none* prompt of 2,165 tokens. *filler* is therefore longer than *degree*, 8% shorter than *clustering* (a conservative control for it), and substantially shorter than *rwse* and *all*, for which it bounds rather than matches the length term.

3.2 Shortcut solvers

A primer’s measured benefit is evidence of graph reasoning only if it exceeds what a solver that never sees the graph can recover from the primer text. Our shortcut solver combines sixteen exact rules, one heuristic, eight fitted rules (fit and scored on disjoint graph sets) and an exact enumeration bound for small graphs. Because it has no access to the graph object, its accuracy on a (task, condition) cell is a content-only lower bound, which we call the *shortcut bar*: it never overstates what the primer text alone permits, but it does not upper-bound model accuracy.

The published GraphQA generator draws graphs of 5–19 nodes, whereas our evaluation corpus has $n=40$, and a bar fit on small graphs does not transfer. Refitting the solver on $n=40$ Erdős–Rényi graphs at the seven densities of our sweep (disjoint 300-graph fit and test sets per density) raises the *edge_existence/none* bar from 0.498 to

Task (5–19 nodes)	none	comp.	clust.	rwse	degree	filler	all
node_count	.064	.076	1.00	1.00	1.00	1.00	1.00
edge_count	.018	.018	.148	.018	1.00	.018	1.00
node_degree	.082	.082	.082	.616	1.00	.082	1.00
connected_nodes	.082	.082	.082	.082	.208	.082	.352
edge_existence	.498	.498	.720	.642	.794	.498	.794
cycle_check	.832	1.00	.832	.832	.946	.832	.946
<i>Task ($n=40$, mean over $p \in \{.10, \dots, .85\}$)</i>							
node_count	1.00	1.00	1.00	1.00	1.00	1.00	1.00
edge_count	.030	.030	.104	.030	1.00	.030	1.00
node_degree	.144	.144	.144	.210	1.00	.144	1.00
connected_nodes	.000	.000	.000	.001	.001	.000	.008
edge_existence	.735	.735	.727	.554	.746	.735	.741
cycle_check	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Table 1: Shortcut bars: accuracy of a solver that reads only the rendered primer, never the graph. Top: 500 graphs from the published generator, used for the held-out arms. Bottom: refit at $n=40$, the size evaluated in the main sweep, averaged over seven densities. **Bold**: the primer content alone determines the answer. The unbolded 1.00 entries at $n=40$ reflect a constant gold answer, not primer content.

0.735, an effect of class imbalance rather than primer content, and reduces the degree-over-none gap on that task from +0.296 to +0.011. We classify a (task, condition) pair as offering a *substitute route* when its bar exceeds the none bar by more than 0.05; at $n=40$ every gain is either at most 0.011 or at least 0.066, so any threshold between these values yields the same split. Under the refit bars (Table 1, bottom), only node_degree (under rwse, degree, all) and edge_count (under clustering, degree, all) offer a route. node_count and cycle_check are excluded from the split: at $n=40$ their gold answers are constant (40, and “yes” at all seven densities), so a blind solver reaches 1.00 under every condition. The published-split bars (Table 1, top) are used for the held-out arms, which answer that corpus.

3.3 Rewiring: isolating structure from length

The filler condition bounds the length confound only approximately (Section 3.1), and comparisons across graph generators cannot isolate a structural quantity: across five generator families at matched (n, m) , the majority-class baseline alone ranges from 0.15 to 1.00. We therefore use a within-instance design. A double-edge swap removes edges (a, b) and (c, d) and adds (a, d) and (c, b) (Fosdick et al., 2018); every endpoint keeps its degree, so the degree sequence, n , m , the gold answer and the prompt length are unchanged, while the triangle count, which the clustering primer reports, changes. We generate low,

base and high triangle-count variants of each graph ($n=40$, mean degree 12, 200 graphs per level; mean clustering coefficient 0.01, 0.31 and 0.73) and evaluate node_degree under none, clustering and filler with qwen3-1.7b, qwen3-1.7b-think and qwen35-2b (Section 4).

The design tests two predictions. If the clustering benefit derives from the stated values, it should vary with triangle count when everything else is fixed. More specifically, local clustering in an Erdős–Rényi graph is approximately the edge probability p , so the primer implicitly states the density, and $\bar{C}(n-1)$ approximates the expected degree. A model that used the primer as a density prior would bias its degree estimates toward $\bar{C}(n-1)$, which equals 0.5, 12.0 and 28.7 across the three levels while the true mean degree stays at 12.

3.4 Answer extraction and measurement validity

Answer extraction errs asymmetrically across conditions, because thinking responses are longer and less regular than direct ones. One failure produced a spurious effect of headline magnitude: the boolean extractor took the last yes/no token, and on cycle_check qwen3-4b often answers correctly and then recites the definition of a cycle (“... a path with no repeated edges or nodes”). Ignoring “no repeated” moved 225 of qwen3-4b’s rows from incorrect to correct and none in the other direction, and removed an apparent condition-dependent collapse; all results use the corrected extractor. A manual check of 50 rows per arm and a corpus-wide comparison found that <think> blocks never reach the extractor unstripped. Scoring truncated generations as incorrect instead of dropping them changes the significance verdict on three qwen3-1.7b cycle_check cells (rwse, clustering, all; Appendix A); the main text drops truncated pairs.

3.5 Significance testing

Every comparison is paired: a condition against none (or another condition) on the same graph and question, with pairs dropped when either side is truncated. The primary test is the exact McNemar test (McNemar, 1947), Benjamini–Hochberg corrected (Benjamini and Hochberg, 1995) within each (arm, task) family of six conditions at $q=0.05$; where we claim that a condition beats both con-

trols, we additionally require significance under a global correction across every test run for that claim. For the relation between effect and baseline (Section 5.4), baseline and effect are estimated from disjoint random halves of each cell’s paired graphs, and the difference between route and no-route cells is tested directly as the interaction in $\Delta \sim \text{baseline} \times \text{route}$, with a bootstrap that resamples whole (arm, task, density) blocks, because conditions within a block share one `none` sample. For each null cell we estimate a minimum detectable effect (MDE): the smallest paired effect that the same test detects with probability 0.8 at $\alpha=0.05$ in simulation (50 replicates, 200 permutations per replicate).

4 Experimental Setup

Tasks. Six GraphQA tasks (Fatemi et al., 2023): node count, edge count, node degree, connected nodes, edge existence and cycle check.

Models. An *arm* is a model checkpoint with a fixed decoding configuration. The main arms are Qwen3-1.7B and Qwen3-4B (Qwen Team, 2025), each with native thinking mode off (`qwen3-1.7b`, `qwen3-4b`) and on (`-think`), chosen because they offer both modes of the same weights at two sizes and moderate cost.

Sweep design. All four arms answer one shared prompt set: Erdős–Rényi graphs (Erdős and Rényi, 1959) with $n=40$ at densities $p \in \{.10, .20, .35, .50\}$, 100 graphs per (density, task) cell, seven conditions and six tasks, for 16,800 prompts per arm. Decoding is greedy with a budget of 2,048 new tokens for plain arms and 8,192 for thinking arms; Section 5.6 reports a budget-matched comparison. A high-density extension covers `node_degree` and `edge_existence` at $p \in \{.65, .75, .85\}$ (100 graphs per cell, all four arms). For the one content effect that survives the controls (Section 5.5), a dedicated `node_degree` sweep with `qwen3-1.7b` uses 400 graphs per density at all seven densities, and a replication uses 400 newly seeded graphs per density at $p \leq .50$.

Held-out arms. Eleven further arms were not used to develop any claim reported here. Eight (Qwen3-8B, Qwen3-14B, Gemma-4-E4B-it and Gemma-4-12B-it (Google DeepMind, 2026), each with and without thinking) answer the published GraphQA split (30 graphs per cell, five tasks, six conditions); three (Qwen3-0.6B with and without

Task	1.7B	1.7B-th.	4B	4B-th.
<code>edge_count</code>	22.4%	81.0%	0.6%	62.2%
<code>cycle_check</code>	6.1%	6.5%	1.0%	1.9%
<code>node_degree</code>	0.0%	1.1%	0.0%	1.6%
<code>connected_nodes</code>	0.0%	0.8%	0.0%	3.5%
<code>edge_existence</code>	0.0%	0.2%	0.1%	0.8%
<code>node_count</code>	0.0%	2.5%	0.0%	0.1%

Table 2: Truncation rate (share of main-sweep generations reaching the output budget: 2,048 tokens plain, 8,192 thinking), by task and arm, pooled over condition and density. **Bold:** above 50%.

thinking, and Qwen3.5-2B (Qwen Team, 2026), `qwen35-2b`) answer a 100-graph probe of `none` and `degree` on all six tasks. The published-split responses were collected before this project’s hypotheses were formed and were analysed earlier for other questions, so we describe these arms as held out rather than pre-registered.

Metrics. Exact match for every task. For `connected_nodes`, set-F1 is 0.96–1.00 in every cell at this graph size and does not discriminate between conditions, so we report exact match. For the integer tasks, mean absolute error is reported in the Appendix.

5 Results

All four arms completed the main sweep (16,800 scored generations each). Unless stated otherwise, Δ is a paired difference against `none`, in percentage points, over instances in which neither condition was truncated.

5.1 Truncation and constant answers

On `edge_count` the thinking arms truncate on 62–81% of instances (Table 2), so surviving pairs are a selected subset that we report but do not interpret; every other task truncates on fewer than 7%. `node_count` (gold = 40) and `cycle_check` (gold = “yes” at every density) have constant answers (Section 3.2); their apparent gains are reported in Table 3 but interpreted as artifacts (Section 5.3), not as graph reasoning.

5.2 The full density sweep

Table 3 (thinking arms in Table 4) pools over the four densities; paired bootstrap intervals for every cell are given in the Appendix. Several cells are significant in one arm without replicating in the other with the same sign and magnitude; Section 5.4 shows why this is expected, since the two

Arm	Task	none	comp.	clust.	rwse	degree	filler	all
qwen3-1.7b	connected_nodes	72.5	+0.8	+1.2	+1.0	+3.5	-6.2	+2.0
	cycle_check	99.7	-0.6	+0.3	-7.5	-12.6	+0.3	-1.1
	edge_count	1.8	-1.4	-0.4	-0.8	+4.0	-0.4	+2.4
	edge_existence	70.5	-8.2	+3.8	+5.3	+4.0	-15.8	+11.5
	node_count	31.5	+21.2	+67.2	+2.0	+6.8	+39.8	+47.0
	node_degree	60.5	+1.8	+3.0	+4.0	+7.5	-0.5	+10.2
qwen3-4b	connected_nodes	86.0	+9.2	+1.0	+0.0	-3.0	-12.5	-2.8
	cycle_check	99.7	-1.3	+0.3	+0.3	-1.5	-0.3	-0.5
	edge_count	2.0	-1.3	-1.3	-1.5	+27.7	+1.2	+3.8
	edge_existence	90.5	-2.8	+4.0	-10.0	-4.3	-7.0	-9.8
	node_count	90.2	+9.0	+8.8	+9.2	+8.0	+9.2	+9.8
	node_degree	99.2	+0.2	-0.5	-2.2	-6.5	-0.5	-8.0

Table 3: Main sweep, plain arms (thinking arms in Table 4), $n=40$, pooled over densities $p \in \{0.10, 0.20, 0.35, 0.50\}$, 100 graphs per (task, primer, density) cell. The none column is absolute accuracy (%); all other columns are paired differences against none in percentage points. Truncated generations excluded pairwise, $n \in [211, 400]$ paired instances/cell. **Bold**: significant, exact McNemar, Benjamini–Hochberg within each (arm, task) family, $q = 0.05$.

Arm	Task	none	comp.	clust.	rwse	degree	filler	all
qwen3-1.7b-think	connected_nodes	79.7	+2.8	+1.5	-0.3	+7.4	+2.0	+6.1
	cycle_check	100.0	+0.0	+0.0	+0.0	-0.7	+0.0	+0.0
	edge_count	86.3	+7.4	-8.7	-4.0	-61.5	+3.4	-8.3
	edge_existence	99.2	-0.8	-1.0	-1.3	-2.3	-0.8	-2.0
	node_count	98.6	+0.9	+1.4	+1.4	+0.8	+1.4	+1.4
	node_degree	76.8	-1.0	+1.0	+2.5	+11.7	-3.5	+11.6
qwen3-4b-think	connected_nodes	98.7	+0.5	+0.0	+0.0	+0.0	+0.8	-0.5
	cycle_check	100.0	+0.0	+0.0	+0.0	-0.8	+0.0	+0.0
	edge_count	97.6	+1.7	+1.7	+0.0	-23.5	-2.0	-41.5
	edge_existence	100.0	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	node_count	99.5	+0.0	+0.5	+0.5	+0.5	+0.5	+0.5
	node_degree	99.7	-0.3	-0.3	-2.1	+0.0	+0.0	-0.8

Table 4: Main sweep, thinking arms; conventions as in Table 3. Both arms are at or near ceiling on most tasks, leaving no headroom for a primer effect. edge_count truncates on 62–81% of rows under thinking (Table 2) and is reported but not interpreted. $n \in [13, 400]$ paired instances/cell. **Bold**: significant, exact McNemar, Benjamini–Hochberg within each (arm, task) family, $q = 0.05$.

model sizes are rarely at matched baseline accuracy on the same cell. Mean absolute error (Appendix Table 13) agrees in sign, except that degree’s +27.7 on edge_count for qwen3-4b leaves it unchanged (39.2 vs. 40.4 edges).

5.3 Length is an active treatment

filler is not inert. It significantly reduces accuracy for qwen3-1.7b on edge_existence (-15.8) and connected_nodes (-6.2), and for qwen3-4b on both (-7.0, -12.5). Over the full seven-density range (qwen3-1.7b-think, node_degree) it costs 6.3 points pooled ($p < 10^{-4}$): negligibly at $p \leq .35$, then 9.5, 9.3, 14.1 and 11.7 points at the four densest levels ($r = +0.953$ with baseline across levels, $p = 0.00088$). The cost of added length thus grows with task difficulty, in the same direction as the benefit of an answer-stating primer (Section 5.4).

Arm	Cond.	hit rate	false alarm
qwen3-1.7b	none	100.0%	40.3%
	filler	100.0%	61.8%
	clustering	100.0%	35.2%
qwen3-4b	none	99.1%	12.6%
	filler	100.0%	22.5%
	clustering	100.0%	7.5%

Table 5: edge_existence, $p=0.50$: hit rate (correct on gold “yes”) and false-alarm rate (“yes” on gold “no”). Accuracy changes almost entirely through the false-alarm rate.

The largest edge_existence effects in Table 3 do not reflect more accurate reading of the graph. Table 5 decomposes accuracy into hit rate and false-alarm rate: the hit rate is 99–100% under every condition for both arms, so accuracy varies only through false alarms, which filler increases (40.3→61.8% and 12.6→22.5%) and

Accuracy under	substitute route		no route	
none	cells	Δ	cells	Δ
0.00–0.25	24	+6.3	24	−0.7
0.25–0.50	6	+8.5	6	+2.0
0.50–0.75	6	+15.5	24	+0.4
0.75–0.95	11	+4.2	33	−3.9
0.95–1.00	34	−9.0	186	−0.4
$r(\Delta, \text{baseline})$	−0.417		−0.050	

Table 6: Mean primer effect (percentage points), binned by none accuracy on the same (arm, task, density, condition) cell of the main sweep, using the $n=40$ route split (Table 1, bottom). Mean baseline is 0.61 for route cells and 0.85 for no-route cells, so the groups are not baseline-matched; the cross-fitted interaction test in the text addresses this.

clustering decreases. A signal-detection analysis (Macmillan and Creelman, 2005), with the log-linear correction for extreme proportions (Hautus, 1995), cannot separate a shift in response criterion from a change in sensitivity here: with the hit rate at ceiling, a change in the false-alarm rate alone moves d' and c together and cannot be attributed to either (Appendix Table 10). We therefore report only that accuracy on this task moves through the false-alarm rate, and do not interpret these effects as changes in graph reading. Under none, qwen3-1.7b’s criterion becomes steadily more liberal with density (c from -0.27 at $p=.10$ to -1.74 at $p=.50$).

node_count’s largest apparent gains are also an artifact. Gold is the constant 40; the dominant none error is “39”, an off-by-one induced by the encoder’s “nodes 0, ..., 39” preamble. qwen3-1.7b answers “39” on 68.5% of instances under none, 61.8% under degree, 28.8% under filler and 1.3% under clustering. Almost any interposed text reduces this error, and none of these changes is a structural effect on a task with a single possible answer.

5.4 Answer-stating primers: route substitution

A cell is an (arm, task, density, condition) combination with at least ten usable pairs. Table 6 splits the 354 main-sweep cells on non-constant tasks by whether the primer offers a substitute route. Route cells gain at low baseline and lose at high baseline ($r=-0.417$, 81 cells); no-route cells do not ($r=-0.050$, 273 cells). Two problems prevent reading this contrast directly: baseline and effect

share sampling noise, and a difference in significance is not a significant difference (Gelman and Stern, 2006). We therefore estimate baseline and effect from disjoint halves of each cell’s graphs and test the route $\times$ baseline interaction, adding the high-density extension (117 route and 381 no-route cells; Figure 1). The interaction is -11.0 points per unit of baseline (95% block-bootstrap CI $[-17.1, -5.7]$, 83 blocks). Within groups, the cross-fitted correlation is $r=-0.34$ for route cells (116 cells, one fold direction, $p=1.6\times 10^{-4}$) and $r=+0.03$ for no-route cells; the other fold direction gives -0.28 and $+0.02$.

Accuracy is bounded, so a cell near 1.0 can only lose and a cell near 0.0 can only gain. On the logit scale, which removes this compression, the interaction persists (-1.46 logits, CI $[-2.11, -0.79]$). Restricted to cells with baseline between 0.10 and 0.90, however, neither group shows a relation (route $r=-0.01$, 78 cells; no-route $r=-0.07$, 187 cells; interaction CI $[-29, +41]$). The evidence therefore supports a contrast between floor and ceiling (a stated answer helps a model that cannot compute it and harms one that already can) rather than a graded dependence that can be detected at intermediate baselines, where our data are sparsest (Appendix Table 8).

The density continuum manipulates difficulty instead of observing it. Holding the arm (qwen3-1.7b-think) and task (node_degree) fixed and raising density from $p=.10$ to $.85$ lowers none accuracy from 0.95 to 0.44, while degree’s benefit rises from $+3.1$ to $+34.2$ points ($r=-0.975$ across levels, $p=1.9\times 10^{-4}$) and filler’s cost grows from negligible to 14.1 points. The two effects run in opposite directions and both grow as the task becomes harder.

Held-out arms and a second family. On the eleven held-out arms the relation reappears where a route exists: $r=-0.674$ over 105 route cells ($p=3.5\times 10^{-15}$) against $r=-0.186$ over 93 no-route cells ($p=0.075$), with an interaction of -24.7 (block-bootstrap CI $[-109, -3.8]$, $p=0.027$, 43 (arm, task) blocks), weaker than on the main sweep but excluding zero. In an ordinary least-squares regression of effect on baseline, log parameter count and thinking mode (classical standard errors, so descriptive only), the baseline coefficient is -30.2 and the parameter coefficient $+0.40$ ($p=0.54$): once baseline is accounted for, scale adds nothing

measurable. Replacing each cell’s own baseline with the other arms’ accuracy on the same cell, a negative control that shares no observation with the effect, removes the relation ($r=+0.113$, $p=0.135$), so the effect tracks a model’s own competence rather than item difficulty. Most held-out arms are near ceiling under none (mean 0.924), and a confirmatory 500-graph run with a pre-specified condition (qwen3-8b, degree) gives +6.67 points ($p=1\times 10^{-4}$).

Renaming nodes. Replacing integer node identifiers with Game-of-Thrones character names throughout the prompt leaves the route effect largely intact: on the published split, degree’s effect correlates at $r=0.933$ between the integer- and name-based sweeps (39 cells), whereas non-route conditions correlate weakly or not at all (clustering $r=0.079$, components $r=0.345$, $rwse\ r=-0.381$). Renaming without a primer changes accuracy by -0.19 points pooled across eight arms (none significant after correction). Because renaming changes identifiers but not the stated degree values, this shows that the effect does not depend on integer labels; it does not distinguish copying the stated value from computing the answer, which requires a primer that states incorrect values (Section 7).

Error analysis. When a primer states the answer and the model still errs, the errors indicate positional copying. Of qwen3-4b’s 29 errors under degree on node_degree, 12 (41.4%) equal the stated degree of the node listed immediately before or after the queried one, against 17.3% expected by chance (binomial $p=.002$); under all, which states the same degree amid three facts per node, the rate is 2 of 35 (5.7%).

5.5 Non-answer content: clustering on node_degree

clustering on node_degree is the only cell in which a primer that does not state the answer was followed up. It was selected after the main sweep and examined in a dedicated sweep and an independent-seed replication (Table 7). Against none, the effect is small when pooled over seven densities (+1.9 points, $p=0.022$) and significant individually only at $p=.35$ (+7.5); it is +0.3 and +3.3 at the two lowest densities, +4.3 at $p=.50$, and within two points of zero at the three highest. The independent-seed replication at $p\leq .50$ confirms a modest gain (+4.2, $p=0.00055$).

Comparison	Δ	n	density	p
clust. vs. none (main)	+3.0	400	.10-.50	0.26
clust. vs. none	+1.9	2800	.10-.85	0.022
filler vs. none	-2.4	2800	.10-.85	0.0013
clust. vs. filler	+4.3	2800	.10-.85	2.4×10^{-8}
clust. vs. none	+7.5	400	.35	0.0089
clust. vs. none (new seed)	+4.2	1600	.10-.50	0.00055

Table 7: qwen3-1.7b, node_degree. All rows except the first come from the dedicated 400-graph-per-density sweep; the last row uses newly seeded graphs. Per-density values are in Appendix Table 9.

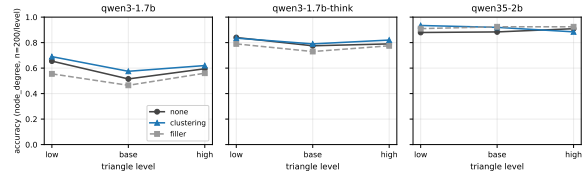


Figure 2: node_degree accuracy by triangle level and condition (200 graphs per level, $n=40$, mean degree 12). Within each arm, the degree sequence, n , m and prompt length are identical across levels; mean clustering is 0.01, 0.31 and 0.73 from low to high.

Against the length-matched filler, the pooled advantage is larger (+4.3, $p=2.4\times 10^{-8}$), but much of it reflects filler’s own cost (-2.4 against none, $p=0.0013$). At $p\geq .65$, where clustering and none do not differ, filler costs 3.5–9.5 points, which accounts for the entire difference there. Only at $p=.35$ and $.50$ does clustering exceed both controls (+6.5 and +6.3 against filler, $p=0.027$ and 0.022). Length does not explain this residual effect, since filler is 40 tokens shorter than clustering. For qwen3-4b, which is at ceiling on this task in the main sweep, the high-density extension gives +11.3 points against none ($p<10^{-3}$) and +9.3 against filler ($p=6\times 10^{-4}$), both significant under the global correction. The effect is absent in qwen3-1.7b-think (Table 4), and qwen3-4b-think is at ceiling on this task throughout.

The rewiring experiment constrains the mechanism but does not identify it (Figure 2). Mean clustering rises from 0.01 to 0.31 to 0.73 across levels by construction, but the benefit does not follow: for qwen3-1.7b, clustering minus none is +3.5, +6.0 and +2.5 points at the three levels, broadly parallel to none’s own level-to-level variation, and for qwen35-2b filler matches or exceeds clustering at every level. The density-prior prediction (Section 3.3) fails as well: it im-

plies that errors under `clustering` should shift upward by up to 17 at the `high` level, but the paired shift in signed error relative to `none` is within 0.3 of zero at every level for all three arms (Appendix Table 12). Neither mechanism we can test accounts for the effect.

5.6 Thinking mode

Thinking brings most cells to ceiling, and where it does not, truncation censors the task (Table 2). Generation length also depends on condition: for `qwen3-1.7b-think`, `degree` raises the non-termination rate on `node_degree` by 3.4 points ($p < 0.001$), producing 130 non-terminating generations outside `edge_count` that do not occur under `none`. `qwen3-4b-think` answers correctly on all 29 instances on which `degree` causes `qwen3-4b` to fail.

Because the two modes had different budgets, we rescored the thinking arms as if capped at the plain arms’ 2,048 tokens; under greedy decoding this is exact, since a trace that finished within 2,048 tokens is unaffected by the lower cap. At equal budget, thinking still outperforms the plain mode on `edge_existence` (97.8 vs. 70.5% for 1.7B; 99.8 vs. 90.5% for 4B) and `node_count`, but underperforms on `cycle_check` (38.0 vs. 90.7%; 62.0 vs. 99.0%) and on `node_degree` for 4B (91.2 vs. 99.3%), and no `edge_count` trace finishes within the budget (Appendix Table 11). Part of the advantage of thinking is therefore an advantage of budget.

5.7 Statistical power

For every null cell in the main sweep we estimate an MDE (Section 3.5). Where it is finite, the median MDE is 5.6 (`qwen3-1.7b`) and 10.0 points (`qwen3-4b`) for gains and 2–3 points for losses. In 10 of 24 null cells for `qwen3-1.7b`, 10 of 20 for `qwen3-4b`, 23 of 33 for `qwen3-1.7b-think` and 33 of 33 for `qwen3-4b-think`, an effect in at least one direction is undetectable at any magnitude because `none` accuracy is at or near 0 or 100%. These nulls reflect ceiling or floor, not evidence of no effect.

6 Discussion

Three regularities account for most of the measured effects. For RQ1, a primer that states the requested quantity acts as a second route to the answer, helping a model at floor and harming one at ceiling;

the contrast survives cross-fitting, a logit transformation, held-out arms, a negative-control difficulty estimate and node renaming, but it is not graded at intermediate baselines, and renaming cannot establish whether the model copies or computes. For RQ2, added text is an active treatment whose cost exceeds most content effects and grows with difficulty, so a primer compared only against `none` confounds content with length. Third, headroom is scarce: thinking saturates most cells, and constant gold answers remove two tasks from consideration. For RQ3, one effect of a primer that does not state the answer survives, `clustering` on `node_degree`, worth four to seven points at intermediate densities for `qwen3-1.7b` and present in `qwen3-4b` below ceiling, but neither mechanism we could test explains it. All claims are behavioural: we measured no internal states of the models.

7 Future Work

Each of the following needs new generations. A *shuffled-value* primer (same template and values, permuted across nodes) is a stricter placebo than *filler*; a *corrupted* primer stating wrong degrees separates copying from computation; and a *single-fact* primer separates interference from route substitution. Varying primer position, encoding and generator family (e.g. Barabási–Albert, in which clustering varies across nodes), matching fillers in tokens to each condition, running the plain arms at 8,192 tokens, and extending the `clustering` sweep across Qwen3 sizes would address the remaining limitations.

8 Conclusion

Structural primers change LLM accuracy on GraphQA mainly by stating the answer, which helps only a model that cannot otherwise produce it, and by lengthening the prompt, which costs accuracy on harder instances; the one surviving effect of non-answer content is small and unexplained. Studies of prompt-level graph augmentations should therefore report baseline accuracy, a length-matched control and a graph-blind solver, and check for constant answers at the evaluated graph size.

Limitations

The main sweep covers two model sizes from one family; the held-out arms extend the range but

on a smaller corpus with fewer conditions and were not pre-registered. The route classification depends on a threshold and on the $n=40$ shortcut refit, and the route relation is not detectable within the 0.10–0.90 baseline window, where data are sparse. All graphs are Erdős–Rényi, in which local clustering is nearly uniform, so the `clustering` primer carries little node-specific information; the rewiring varies triangle count within this family but does not test other families. The length control matches `clustering` to within 8% in tokens but is much shorter than `rwse` and `all`. Every prompt uses one encoding (`incident`), one template and one primer position, with greedy decoding. The `clustering` cell was selected after the main sweep, and several results rest on 400 or fewer instances per cell; an MDE is reported for every null result, but it does not replace a larger sample.

Ethics Statement

This work evaluates open-weight language models on synthetically generated Erdős–Rényi graphs. It involves no human subjects, no personal or identifying data and no annotation labour. We see no direct risk of harm from the findings; the principal cost is the GPU compute for approximately 200,000 generations.

References

- Yoav Benjamini and Yosef Hochberg. 1995. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1):289–300.
- Victor Chernozhukov, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins. 2018. Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68.
- Rotem Dror, Gili Baumer, Segev Shlomov, and Roi Reichart. 2018. The hitchhiker’s guide to testing statistical significance in natural language processing. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 1383–1392.
- Vijay Prakash Dwivedi, Anh Tuan Luu, Thomas Laurent, Yoshua Bengio, and Xavier Bresson. 2022. Graph neural networks with learnable structural and positional representations. In *International Conference on Learning Representations (ICLR)*.
- Nouha Dziri, Ximing Lu, Melanie Sclar, Xiang Lorraine Li, Liwei Jiang, Bill Yuchen Lin, Peter West, Chandra Bhagavatula, Ronan Le Bras, Jena D. Hwang, Soumya Sanyal, Sean Welleck, Xiang Ren, Allyson Ettinger, Zaid Harchaoui, and Yejin Choi. 2023. Faith and fate: Limits of transformers on compositionality. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36.
- Paul Erdős and Alfréd Rényi. 1959. On random graphs i. *Publicationes Mathematicae Debrecen*, 6:290–297.
- Bahare Fatemi, Jonathan Halcrow, and Bryan Perozzi. 2023. Talk like a graph: Encoding graphs for large language models. *arXiv preprint arXiv:2310.04560*.
- Bailey K. Fosdick, Daniel B. Larremore, Joel Nishimura, and Johan Ugander. 2018. Configuring random graph models with fixed degree sequences. *SIAM Review*, 60(2):315–355.
- Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. 2020. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2(11):665–673.
- Andrew Gelman and Hal Stern. 2006. The difference between “significant” and “not significant” is not itself statistically significant. *The American Statistician*, 60(4):328–331.
- Google DeepMind. 2026. Gemma 4 model cards (gemma-4-E4B-it, gemma-4-12B-it). <https://huggingface.co/google>.
- Jiayan Guo, Lun Du, and Hengyu Liu. 2023. Gpt4graph: Can large language models understand graph structured data? an empirical evaluation and benchmarking. *arXiv preprint arXiv:2305.15066*.
- Michael J. Hautus. 1995. Corrections for extreme proportions and their biasing effects on estimated values of d' . *Behavior Research Methods, Instruments, & Computers*, 27(1):46–51.
- Bowen Jin, Gang Liu, Chi Han, Meng Jiang, Heng Ji, and Jiawei Han. 2024. Large language models on graphs: A comprehensive survey. *IEEE Transactions on Knowledge and Data Engineering*, 36(12):8622–8642.
- Greg Kamradt. 2023. Needle in a haystack – pressure testing llms. https://github.com/gkamradt/LLMTest_NeedleInAHaystack.
- Mosh Levy, Alon Jacoby, and Yoav Goldberg. 2024. Same task, more tokens: the impact of input length on the reasoning performance of large language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 15339–15353.
- Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. 2024. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12:157–173.

Neil A. Macmillan and C. Douglas Creelman. 2005. *Detection Theory: A User’s Guide*, 2nd edition. Lawrence Erlbaum Associates.

Quinn McNemar. 1947. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2):153–157.

Bryan Perozzi, Bahare Fatemi, Dustin Zelle, Anton Tsitsulin, Mehran Kazemi, Rami Al-Rfou, and Jonathan Halcrow. 2024. Let your graph do the talking: Encoding structured data for LLMs. *arXiv preprint arXiv:2402.05862*.

Qwen Team. 2025. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*.

Qwen Team. 2026. Qwen3.5-2B model card. <https://huggingface.co/Qwen/Qwen3.5-2B>.

Freda Shi, Xinyun Chen, Kanishka Misra, Nathan Scales, David Dohan, Ed H. Chi, Nathanael Schärli, and Denny Zhou. 2023. Large language models can be easily distracted by irrelevant context. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, volume 202 of *PMLR*, pages 31210–31227.

Heng Wang, Shangbin Feng, Tianxing He, Zhaoxuan Tan, Xiaochuang Han, and Yulia Tsvetkov. 2023. Can language models solve graph problems in natural language? In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. 2022. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35.

A Supplementary material

A.1 Example prompt

A clustering-condition node_degree prompt at $p=0.10$ (elisions marked ...). Other conditions replace the first paragraph; none omits it.

Node 0 has clustering coefficient 0.20. Node 1 has clustering coefficient 0.00. ... Node 39 has clustering coefficient 0.00.

G describes a graph among nodes 0, 1, 2, ..., 38, and 39.

In this graph:

Node 0 is connected to nodes 2, 6, 8, 12, 35. Node 1 is connected to nodes 5, 20.

...

Node 39 is connected to nodes 15, 16, 18, 20.

Q: What is the degree of node 16?

A:

A.2 Robustness analyses

Estimate	r_{route}	k	r_{none}	k	interaction	[95% CI]
naive, pp	-0.35	117	-0.01	381	-12.5	[-19.5, -6.3]
cross-fit (one direction), pp	-0.34	116	+0.03	381	-12.1	[-17.9, -7.0]
cross-fit (both), pp	-0.31	232	+0.02	762	-11.0	[-17.1, -5.7]
cross-fit (both), logit	-0.40	232	+0.04	762	-1.35	[-1.93, -0.77]
baseline .10-.90, pp	-0.01	78	-0.07	187	+3.0	[-29.1, +41.1]
baseline .10-.90, logit	+0.02	78	-0.05	187	+0.22	[-1.36, +1.94]
held-out, pp	-0.67	105	-0.19	93	-24.7	[-109, -3.8]

Table 8: Route $\times$ baseline relation, main sweep plus high-density extension (held-out: published split and probe). “Both” counts each cell once per fold direction; k for the window rows counts points. Interaction CIs resample (arm, task, density) blocks, or (arm, task) for held-out arms.

p	clust.-none	filler-none	clust.-filler
.10	+0.3	+3.0*	-2.8*
.20	+3.3	+0.5	+2.8
.35	+7.5*	+1.0	+6.5*
.50	+4.3	-2.0	+6.3*
.65	-2.0	-9.5*	+7.5*
.75	+0.5	-6.3*	+6.8*
.85	-0.5	-3.5*	+3.0*

Table 9: qwen3-1.7b, node_degree, dedicated sweep, 400 graphs per density. *: McNemar $p < 0.05$ (uncorrected).

Cond.	qwen3-1.7b		qwen3-4b	
	$\Delta d'$	Δc	$\Delta d'$	Δc
components	-0.36 [-0.56, -0.21]	-0.18	+0.07 [-0.24, +0.33]	-0.10
clustering	+0.21 [+0.03, +0.41]	+0.11	+0.43 [+0.12, +0.75]	+0.08
rwse	+0.07 [-0.19, +0.43]	+0.28	-0.38 [-0.68, -0.15]	-0.32
degree	+0.15 [-0.05, +0.41]	+0.19	-0.04 [-0.33, +0.21]	-0.15
filler	-0.93 [-1.16, -0.71]	-0.46	-0.25 [-0.54, +0.00]	-0.25
all	+0.55 [+0.35, +0.77]	+0.27	-0.36 [-0.71, -0.11]	-0.31

Table 10: edge_existence, main sweep: change in sensitivity d' and criterion c against none, averaged over the four densities (log-linear correction; 95% bootstrap CI for $\Delta d'$). With the hit rate at ceiling, d' and c are not separately identifiable.

Task	1.7B, acc. under none (%)			4B, acc. under none (%)		
	plain	think	think@2048	plain	think	think@2048
connected_nodes	72.5	78.7	67.2	86.0	95.5	89.0
cycle_check	90.7	97.0	38.0	99.0	99.3	62.0
edge_count	1.3	15.8	0.0	2.0	20.0	0.0
edge_existence	70.5	99.0	97.8	90.5	99.8	99.8
node_count	31.5	91.0	86.0	90.2	99.5	94.8
node_degree	60.5	76.2	67.7	99.3	97.0	91.2

Table 11: Budget-matched comparison. Plain arms use 2,048 tokens; thinking arms 8,192 (“think”) or rescored at 2,048 (“think@2048”). Truncated generations are scored incorrect here, so values differ from the dropped-pair convention of Tables 3 and 4.

Arm	low	base	high
predicted shift	-11.5	0.0	+16.7
qwen3-1.7b	+0.03	+0.28	-0.02
qwen3-1.7b-think	+0.01	+0.04	+0.01
qwen35-2b	-0.29	-0.26	-0.16

Table 12: Density-prior test on the rewiring corpus: mean paired shift in signed node_degree error, clustering minus none (errors clipped at ± 20 , 200 graphs per level). “Predicted” is the shift if answers moved fully to $\bar{C}(n-1)$.

Arm	Task	none	comp.	clust.	rwse	degree	filler	all
1.7B	node_count	0.69	0.47	0.01	0.67	0.62	0.29	0.22
	edge_count	133.4	98.1	105.7	79.0	47.7	181.2	94.2
	node_degree	0.94	0.91	0.78	0.91	0.67	1.27	0.91
4B	node_count	0.10	0.01	0.01	0.01	0.02	0.01	0.00
	edge_count	39.2	40.1	43.4	56.0	40.4	29.3	28.1
	node_degree	0.02	0.01	0.02	0.05	0.14	0.02	0.49

Table 13: Mean absolute error on the integer tasks, plain arms, main sweep (non-truncated rows).

A.3 Per-cell results

Paired effect sizes for every (arm, task, primer) cell: Δ in percentage points against none, a 95% paired-bootstrap interval, the permutation p -value, and Δ under the alternative convention that scores truncated generations incorrect rather than dropping the pair (Section 3.4). n is the number of pairs in which neither arm truncated.

Task	Primer	n	Δ	95% CI	perm p	zeroed
connected_nodes	components	400	+0.8	[-2.8, +4.5]	0.7861	+0.8
	clustering	400	+1.2	[-3.0, +5.8]	0.6458	+1.2
	rwse	400	+1.0	[-3.5, +5.2]	0.7304	+1.0
	degree	400	+3.5	[-0.5, +7.5]	0.1239	+3.5
	filler	400	-6.2	[-10.2, -2.2]	0.0040	-6.2
	all	399	+2.0	[-2.8, +6.8]	0.4644	+2.0
cycle_check	components	343	-0.6	[-1.5, +0.0]	0.5033	+2.0
	clustering	362	+0.3	[+0.0, +0.8]	1.0000	+8.5
	rwse	358	-7.5	[-10.3, -4.7]	0.0001	-0.2
	degree	301	-12.6	[-16.6, -9.0]	0.0001	-17.5
	filler	341	+0.3	[+0.0, +0.9]	1.0000	+3.0
	all	362	-1.1	[-2.5, +0.0]	0.2210	+7.0
edge_count	components	211	-1.4	[-3.8, +0.5]	0.3783	-1.0
	clustering	238	-0.4	[-2.5, +1.7]	1.0000	-0.5
	rwse	239	-0.8	[-2.9, +1.3]	0.6837	-0.8
	degree	253	+4.0	[+0.8, +7.1]	0.0179	+3.0
	filler	224	-0.4	[-2.2, +1.3]	1.0000	-0.5
	all	211	+2.4	[-0.5, +5.7]	0.2269	+0.8
edge_existence	components	400	-8.2	[-11.8, -4.8]	0.0001	-8.2
	clustering	400	+3.8	[-0.5, +8.0]	0.1199	+3.8
	rwse	399	+5.3	[+0.8, +9.8]	0.0321	+5.0
	degree	400	+4.0	[-0.5, +8.2]	0.0957	+4.0
	filler	400	-15.8	[-20.0, -11.8]	0.0001	-15.8
	all	400	+11.5	[+6.8, +16.2]	0.0001	+11.5
node_count	components	400	+21.2	[+15.8, +26.8]	0.0001	+21.2
	clustering	400	+67.2	[+62.7, +71.8]	0.0001	+67.2
	rwse	400	+2.0	[-4.2, +8.5]	0.5871	+2.0
	degree	400	+6.8	[-3.0, +10.5]	0.0008	+6.8
	filler	400	+39.8	[+35.0, +44.8]	0.0001	+39.8
	all	400	+47.0	[+42.0, +52.0]	0.0001	+47.0
node_degree	components	400	+1.8	[-2.0, +5.8]	0.4599	+1.8
	clustering	400	+3.0	[-1.8, +7.8]	0.2577	+3.0
	rwse	399	+4.0	[-0.5, +8.5]	0.1104	+4.0
	degree	400	+7.5	[-3.0, +12.0]	0.0018	+7.5
	filler	400	-0.5	[-4.5, +3.5]	0.9028	-0.5
	all	400	+10.2	[+5.2, +15.2]	0.0003	+10.2

Table 14: qwen3-1.7b: all cells.

Task	Primer	n	Δ	95% CI	perm p	zeroed
connected_nodes	components	392	+2.8	[-0.5, +6.1]	0.1268	+2.5
	clustering	394	+1.5	[-2.5, +5.6]	0.5384	+2.0
	rwse	393	-0.3	[-4.1, +3.6]	1.0000	+0.2
	degree	390	+7.4	[+3.6, +11.3]	0.0001	+6.2
	filler	394	+2.0	[-1.5, +5.6]	0.3275	+2.5
	all	392	+6.1	[+2.6, +9.9]	0.0027	+6.2
cycle_check	components	346	+0.0	[+0.0, +0.0]	1.0000	-7.5
	clustering	376	+0.0	[+0.0, +0.0]	1.0000	-0.5
	rwse	386	+0.0	[+0.0, +0.0]	1.0000	+2.2
	degree	295	-0.7	[-1.7, +0.0]	0.5036	-21.8
	filler	388	+0.0	[+0.0, +0.0]	1.0000	+3.0
	all	373	+0.0	[+0.0, +0.0]	1.0000	-0.8
edge_count	components	27	+7.4	[+0.0, +18.5]	0.5037	-0.8
	clustering	23	-8.7	[-26.1, +8.7]	0.6251	-1.2
	rwse	25	-4.0	[-12.0, +0.0]	1.0000	-3.2
	degree	13	-61.5	[-84.6, -38.5]	0.0102	-3.2
	filler	29	+3.4	[-6.9, +13.8]	1.0000	-1.5
	all	24	-8.3	[-29.2, +12.5]	0.6794	+5.0
edge_existence	components	398	-0.8	[-2.3, +0.8]	0.5090	-0.8
	clustering	398	-1.0	[-2.5, +0.3]	0.2911	-1.0
	rwse	398	-1.3	[-3.0, +0.3]	0.2274	-1.2
	degree	397	-2.3	[-4.3, +0.5]	0.0249	-2.5
	filler	399	-0.8	[-2.3, +0.8]	0.5051	-0.5
	all	399	-2.0	[-3.8, -0.5]	0.0374	-1.8
node_count	components	342	+0.9	[-0.3, +2.3]	0.3758	+1.5
	clustering	369	+1.4	[+0.3, +2.7]	0.0597	+9.0
	rwse	369	+1.4	[+0.3, +2.7]	0.0597	+9.0
	degree	360	+0.8	[-0.6, +2.2]	0.4512	+5.5
	filler	369	+1.4	[+0.3, +2.7]	0.0597	+9.0
	all	369	+1.4	[+0.3, +2.7]	0.0597	+9.0
node_degree	components	395	-1.0	[-4.3, +2.3]	0.6543	-1.0
	clustering	396	+1.0	[-2.8, +4.8]	0.7010	+1.5
	rwse	394	+2.5	[-1.5, +6.9]	0.2898	+2.5
	degree	376	+11.7	[+7.4, +16.0]	0.0001	+8.8
	filler	396	-3.5	[-7.3, +0.3]	0.0954	-3.0
	all	396	+11.6	[+7.3, +15.9]	0.0001	+12.2

Table 15: qwen3-1.7b-think: all cells.

Task	Primer	n	Δ	95% CI	perm p	zeroed
connected_nodes	components	400	+9.2	[+6.0, +12.8]	0.0001	+9.2
	clustering	400	+1.0	[-3.0, +4.8]	0.6893	+1.0
	rwse	400	+0.0	[-4.0, +4.0]	1.0000	+0.0
	degree	400	-3.0	[-7.2, +1.2]	0.1882	-3.0
	filler	400	-12.5	[-17.0, -8.0]	0.0001	-12.5
	all	400	-2.8	[-7.0, +1.5]	0.2511	-2.8
cycle_check	components	391	-1.3	[-2.6, +0.0]	0.1318	-2.0
	clustering	395	+0.3	[+0.0, +0.8]	1.0000	+0.5
	rwse	392	+0.3	[+0.0, +0.8]	1.0000	-0.2
	degree	391	-1.5	[-2.8, -0.5]	0.0312	-2.2
	filler	392	-0.3	[-0.8, +0.0]	1.0000	-0.5
	all	395	-0.5	[-1.5, +0.5]	0.6269	-0.2
edge_count	components	397	-1.3	[-2.8, +0.3]	0.1756	-1.2
	clustering	399	-1.3	[-3.0, +0.3]	0.2298	-1.2
	rwse	398	-1.5	[-3.0, -0.3]	0.0668	-1.5
	degree	393	+27.7	[+23.2, +32.6]	0.0001	+27.3
	filler	400	+1.2	[-0.5, +3.2]	0.3012	+1.2
	all	397	+3.8	[+1.3, +6.3]	0.0049	+3.8
edge_existence	components	400	-2.8	[-6.0, +0.5]	0.1254	-2.8
	clustering	400	+4.0	[+1.0, +7.0]	0.0168	+4.0
	rwse	400	-10.0	[-14.0, -6.2]	0.0001	-10.0
	degree	398	-4.3	[-7.8, -1.0]	0.0200	-4.8
	filler	400	-7.0	[-10.8, -3.5]	0.0002	-7.0
	all	399	-9.8	[-13.8, -6.0]	0.0001	-10.0
node_count	components	400	+9.0	[+6.2, +12.0]	0.0001	+9.0
	clustering	400	+8.8	[+5.8, +11.8]	0.0001	+8.8
	rwse	400	+9.2	[+6.2, +12.2]	0.0001	+9.2
	degree	400	+8.0	[+5.0, +11.2]	0.0001	+8.0
	filler	400	+9.2	[+6.5, +12.2]	0.0001	+9.2
	all	400	+9.8	[+7.0, +12.8]	0.0001	+9.8
node_degree	components	400	+0.2	[+0.0, +0.8]	1.0000	+0.2
	clustering	400	-0.5	[-2.0, +0.8]	0.7246	-0.5
	rwse	400	-2.2	[-4.0, -0.5]	0.0240	-2.2
	degree	400	-6.5	[-9.2, -3.8]	0.0001	-6.5
	filler	400	-0.5	[-2.0, +0.8]	0.7232	-0.5
	all	400	-8.0	[-10.8, -5.2]	0.0001	-8.0

Table 16: qwen3-4b: all cells.

Task	Primer	n	Δ	95% CI	perm p	zeroed
connected_nodes	components	377	+0.5	[-0.5, +1.6]	0.6249	+0.5
	clustering	373	+0.0	[-1.3, +1.3]	1.0000	-0.8
	rwse	379	+0.0	[-1.3, +1.6]	1.0000	+0.5
	degree	376	+0.0	[-1.6, +1.6]	1.0000	+0.5
	filler	379	+0.8	[-0.3, +2.1]	0.3787	+1.2
	all	375	-0.5	[-2.1, +0.8]	0.7306	-1.2
cycle_check	components	381	+0.0	[+0.0, +0.0]	1.0000	-3.2
	clustering	394	+0.0	[+0.0, +0.0]	1.0000	+0.0
	rwse	395	+0.0	[+0.0, +0.0]	1.0000	+0.2
	degree	375	-0.8	[-1.9, +0.0]	0.2514	-5.5
	filler	396	+0.0	[+0.0, +0.0]	1.0000	+0.5
	all	392	+0.0	[+0.0, +0.0]	1.0000	-0.5
edge_count	components	60	+1.7	[+0.0, +5.0]	1.0000	+24.8
	clustering	59	+1.7	[-3.4, +6.8]	1.0000	+24.8
	rwse	49	+0.0	[-6.1, +6.1]	1.0000	+10.2
	degree	34	-23.5	[-38.2, -8.8]	0.0073	+8.5
	filler	50	-2.0	[-6.0, +0.0]	1.0000	+12.2
	all	41	-41.5	[-56.1, -26.8]	0.0002	+1.8
edge_existence	components	396	+0.0	[+0.0, +0.0]	1.0000	-0.5
	clustering	396	+0.0	[+0.0, +0.0]	1.0000	-0.8
	rwse	393	+0.0	[+0.0, +0.0]	1.0000	-1.2
	degree	398	+0.0	[+0.0, +0.0]	1.0000	+0.0
	filler	399	+0.0	[+0.0, +0.0]	1.0000	+0.0
	all	394	+0.0	[+0.0, +0.0]	1.0000	-1.0
node_count	components	400	+0.0	[-1.0, +1.0]	1.0000	+0.0
	clustering	399	+0.5	[+0.0, +1.3]	0.4936	+0.2
	rwse	399	+0.5	[+0.0, +1.3]	0.5005	+0.2
	degree	399	+0.5	[+0.0, +1.3]	0.5005	+0.2
	filler	400	+0.5	[+0.0, +1.2]	0.4870	+0.5
	all	400	+0.5	[+0.0, +1.2]	0.4870	+0.5
node_degree	components	380	-0.3	[-0.8, +0.0]	1.0000	+0.2
	clustering	386	-0.3	[-1.0, +0.5]	1.0000	+1.8
	rwse	381	-2.1	[-3.7, -0.8]	0.0072	-1.2
	degree	388	+0.0	[-0.8, +0.8]	1.0000	+2.5
	filler	386	+0.0	[+0.0, +0.0]	1.0000	+2.0
	all	383	-0.8	[-1.8, +0.0]	0.2466	+0.5

Table 17: qwen3-4b-think: all cells.